VII SEMESTER (OPEN ELECTIVE III)	L	T	P	C
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20EE7O01 - ENERGY AUDITING, CONSERVATION AND MANAGEMENT				

COURSE OUTCOMES:

After successful completion of this course, students should be able to:

CO1: Understand the principles of energy audit.

CO2: Explain the role of Energy Manager and Energy Management program.

CO3: Design a energy efficient motors and good lighting system

CO4: Evaluate the methods to improve the power factor.

CO5: Estimate the computational techniques with regard to economic aspects.

UNIT-I: PRINCIPLES OF ENERGY AUDIT

Energy audit- definitions - concept - types of audit - energy index - cost index - pie charts – Sankey diagrams and load profiles - Energy conservation schemes and energy saving potential - Energy audit of industries- energy saving potential - energy audit of process industry - thermal power station - building energy audit– Numerical problems.

UNIT-II: ENERGY MANAGEMENT

Principles of energy management - organizing energy management program - initiating – planning - controlling - promoting - monitoring - reporting. Energy manager - qualities and functions – language - Questionnaire – check list for top management.

UNIT-III: ENERGY EFFICIENT MOTORS AND LIGHTING

Energy Efficient Motors: Energy efficient motors - factors affecting efficiency - loss distribution - constructional details - characteristics – variable speed - RMS - voltage variation-voltage unbalance-over motoring-motor energy audit.

Lighting : lighting system design and practice - lighting control - lighting energy audit.

UNIT-IV: POWER FACTOR IMPROVEMENT AND ENERGY INSTRUMENTS

Power factor – methods of improvement - location of capacitors - Power factor with non-linear loads - effect of harmonics on p.f - p.f motor controllers – Energy Instruments- watt meter - data loggers - thermocouples - pyrometers - lux meters - tongue testers.

UNIT-V: ECONOMIC ASPECTS AND COMPUTATION

ECONOMIC ASPECTS: Economics Analysis depreciation Methods - time value of money - rate of return - present worth method - replacement analysis - lifecycle costing analysis.

COMPUTATION ASPECTS: Calculation of simple payback method - net present value method- Power factor correction - lighting – Applications of life cycle costing analysis - return on investment.

Text Books:

1. Energy management by W.R.Murphy & G.Mckay Butter worth - Heinemann publications - 1982.
2. Energy management hand book by W.CTurner - John wiley and sons - 1982.

Reference Books:

1. Energy efficient electric motors by John.C.Andreas - Marcel Dekker Inc Ltd-2nd edition – 1995.
2. Energy management by Paul o' Callaghan - Mc-graw Hill Book company-1st edition – 1998.
   Energy management and good lighting practice : fuel efficiency- booklet12